

Improved Upper Bounds for q-ary Covering Codes with Reproducible and Proof-Carrying Verification

Research artifact: `thiagopatzdorf/james-theorems`

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Abstract

We archive eight recovered explicit covering-code constructions and independently verify their complete ambient spaces by an exact coordinate-wise min-plus transform and multi-source breadth-first search. The principal construction has 1351 distinct words over an alphabet of size seven, length nine and covering radius exactly four. It establishes computational evidence for $K_7(9, 4) \leq 1351$, improving the verified bound 1475 found in our literature audit by 124 words. Its original-order bytes are frozen by SHA-256. A packed selector certificate gives an independently checked radius-four witness for every one of the $7^9 = 40\,353\,607$ ambient words. A separately pinned Lean project proves a small explicit construction and the principal code's cardinality bound, using Florath's existing theory. The principal coverage certificate has not been replayed in Lean. Validity, formal verification and global novelty are reported separately; no best-known or world-record claim is made.

1 Introduction

The purpose is a checkable construction, independent of confidence in a search agent or its logs. Recovered codewords, canonical bytes, executable results and kernel-checked proofs are distinct objects. No new construction search or optimization was performed during this audit. The principal result is classified `COMPUTATIONALLY_VERIFIED`; its novelty classification is `CANDIDATE_NEW_UPPER_BOUND` relative to the audited comparison.

2 Definitions

Let $A_q = \{0, \dots, q-1\}$. For $x, c \in A_q^n$ set $d_H(x, c) = |\{i : x_i \neq c_i\}|$. A finite $C \subseteq A_q^n$ covers at radius R if $\forall x \exists c \in C : d_H(x, c) \leq R$. Write $K_q(n, R)$ for the minimum size of such a code. Its ball volume and sphere ratio are

$$V_q(n, R) = \sum_{i=0}^R \binom{n}{i} (q-1)^i, \quad S = q^n / V_q(n, R), \quad K_q(n, R) \geq \lceil S \rceil.$$

If $L \leq K_q(n, R) \leq U$, the normalized quantity $\alpha = K_q(n, R)/S$ satisfies $L/S \leq \alpha \leq U/S$; it is not a single computed value unless $K_q(n, R)$ is known exactly.

3 Previous bounds

Marosi's current arXiv version [1], v3 dated 2 September 2026, gives an explicit 1475-word construction for the principal cell. We replayed that file using our BFS checker and the published

independent dilation checker; both cover the entire space, and the exact radius is four. Thus 1475 is the best verified prior bound found in this audit. Kéri’s 2011 tables [2] and Gijswijt–Polak’s v2 [3] supply documentary comparisons for the remaining cells. The tables below distinguish documentary bounds from independently replayed ones. Six exact rational lower-bound certificates were checked with Marosi’s standalone checker; these give 241 for the principal cell, weaker than the published 264. The audit does not relabel 264 as a replayed lower-bound certificate.

4 Construction method

The files were recovered from `thiagopatzdorf/fabrica-de-sites`, branch `research/preco-da-impossibilidade`, commit `e7f3ed9c52e6992eaa244fc75a04ed4abed89fd1`. The archived report describes the principal as three translates of a $[9, 3]_7$ linear kernel (1029 words) with 322 patch words. The three-coset decomposition and generator matrix have now been independently checked against the actual prefix of the recovered file. The patch search was not replayed: the original generator revision and seed are unknown. The construction proved computationally is the actual recovered file, not this narrative. The secondary files are reported as linear codes or unions of cosets, with patches in the seven-ary length-eight case. Their generation claims are archived, not assumed by the checkers.

The recovered LF files already satisfy the canonical format: one length- n word per line, symbols $0, \dots, q - 1$, final newline, original order preserved. The principal LF hash is

6d1b0e1abb8079df06a28d5301607d3d5247e0f2005d72e13f695ce26f6e6b52.

Converting these bytes to CRLF reproduces the old raw-file hash, while sorting LF lines reproduces the old certificate hash. These comparisons support archive consistency but neither sorted nor reconstructed bytes replace the frozen code.

5 Verification protocol

Verifier A includes a literal exhaustive nearest-codeword oracle for small tests and a NumPy min-plus transform for large cases. Initialize $D(c) = 0$ at codewords and $D(x) = n + 1$ elsewhere. For each coordinate i apply

$$D_{\text{new}}(x) = \min\{D_{\text{old}}(x), 1 + \min_{a \in A_q} D_{\text{old}}(x_1, \dots, a, \dots, x_n)\}.$$

This is the exact coordinate cost, zero for equality and one for a replacement. Separability of the sum of coordinate costs proves that after all n transforms, $D(x) = \min_{c \in C} d_H(x, c)$. The sentinel is above all realizable distances; intermediate uint8 values are at most $n + 2 \leq 34$.

Verifier B, implemented independently in C++, seeds a breadth-first search of the Hamming graph at all codewords. Each edge changes exactly one coordinate, so graph distance equals Hamming distance. It visits all q^n packed integer vertices, independent of the requested radius. It uses checked integer capacity, signed neighbor arithmetic and a complete-queue invariant. The algorithms share neither their main traversal nor their hash implementation.

All 255 nonempty binary length-three codes at all four radii are cross-checked between the literal oracle, transform and BFS; additional seeded tests cover $q = 3, 4, 5, 7, 10$. Parser attacks include duplicates, BOM, CRLF, missing newline, truncated words, invalid digits and control bytes. Reports include raw and unique cardinality, hash, ambient count, uncovered count and exact covering radius.

The selector certificate stores one uint16 little-endian original-order codeword index per ambient integer in big-endian base- q order. Its payload is 80 707 214 bytes, stored in deterministic gzip. A separate streaming checker reconstructs every ambient word and checks its selected word’s literal Hamming distance. Truncated, extended, out-of-range and mathematically incorrect witnesses are rejected even when their hashes are updated.

A compact partition certificate is also generated and checked. Let L be the 343-word linear span described in the archived matrix, and C_0 the three-coset 1029-word prefix of the actual file. The checker confirms those complete cosets and the bijective projection of L to its first three coordinates. Coordinate translation modulo seven is a Hamming isometry and $C_0 + L = C_0$. Each word has exactly one normalized representative with prefix 000. A checked witness in C_0 for that representative transports to every word of its orbit. Marked representatives instead have direct witnesses for every word in the orbit. The partition therefore exhausts the entire space without a new construction. There are $7^6 = 117649$ representatives, with 27 exceptional orbits containing 9261 points. The quotient and exception uint16 payloads total 253820 bytes; the checker verifies 40 344 346 points by symmetry and 9261 explicitly. This compact route is preferred for future kernel replay; both it and the original flat full-space certificate are retained. Their external PASS reports are not substituted for a Lean theorem.

6 Verified and candidate upper bounds

Theorem 1 (Computationally verified construction) *The frozen 1351-word file has 1351 distinct elements of A_7^9 and satisfies $\max_x \min_{c \in C} d_H(x, c) = 4$. Consequently $K_7(9, 4) \leq 1351$.*

The evidence for this theorem is exhaustive execution and the explicit object; it is not yet a Lean proof. Both checkers report 40 353 607 covered words and zero uncovered. Every secondary code likewise passes both checkers, with no duplicates and exact radius equal to its stated R .

q	n	R	Candidate	Rechecked LB	Rechecked UB	Published interval
7	9	4	1351	241	1475	264–1475
4	10	4	192	51	–	62–208
5	7	2	500	236	–	236–525
5	9	3	1250	354	–	354–1275
5	10	4	625	158	–	177–875
5	9	4	250	62	–	64–255
5	9	5	50	16	–	19–55
7	8	3	1893	471	–	471–2337

Here “Rechecked LB” includes sphere counting or independently validated rational certificates. “Rechecked UB” is a replayed *prior* code, hence absent for secondary cells. Candidate cardinalities in this table are all computationally verified; the word candidate qualifies novelty/formal status, not a missing file. Machine-generated exact rational α intervals are in `bounds.json`.

7 Lean formal verification

We pin Florath’s library [4] at 460df105545c2d6b04ba71f29de6b56dbda92825, using its toolchain Lean 4.30.0-rc2. `QaryWord` is `Fin n → Fin q`; `ExplicitQaryUpper` supplies a `Finset`, cardinality proof and `CoversFinset` proof, and `toUpper` yields `QaryKUpper`. No covering-code foundations are reimplemented.

A concurrent research-branch commit also supplies the classical sphere lower bound $K_7(9, 4) \geq 221$ via the upstream sphere-covering theorem; this source is preserved and included in the final build. The actual build proves $K_2(3, 1) \leq 2$ using `{000, 111}` and `covering_decide+kernel`, plus generic selector lemmas and a cardinality bound for the 1351-word principal Finset loaded in original order. There are zero `sorry`, `admit` or new axiom declarations. Printed dependencies are only `propext`, `Classical.choice` and `Quot.sound`. Native mode is not enabled; upstream native mode overrides the kernel flag if explicitly enabled. The principal coverage theorem is absent. A generic theorem requiring an `ExplicitQaryUpper` hypothesis is not a proof that such an object exists. The modular-translation isometry, explicit matrix normalization, injectivity, 343-word coset cardinality and coset transport are now proved in Lean. Decoding the actual compact payloads, linking the cosets to the original Finset and reflecting the finite witness checks remain work.

8 Reproducibility

Install Python 3.12, NumPy 2.3.5, g++ and OpenSSL headers; install elan for the pinned nested project, and TeX for the paper. From the repository root:

```
make covering-fast
make verify-all-computational
make covering-formal
make verify-q7-9-4-1351
make paper
```

The principal command checks hashes, both exhaustive verifiers, the packed certificate and the actual Lean library build. It deliberately returns nonzero until an unconditional principal theorem is replayed; successful computational reproduction is available separately via `makeverify-computational`. PR CI uses fast checks and the small/formal library; exhaustive checking is manual, not run on every push. CI results are not claimed until the workflow executes. The original cache download failed HTTP 403/429; the actual Lean dependencies were built from source. `lake update` succeeds with the upstream-supported `MATHLIB_NO_CACHE_ON_UPDATE=1` setting.

9 Limitations

The computational trusted base includes Python/NumPy, C++/OpenSSL, their compilers and the operating system; testing is not a proof of all executable semantics. The principal selector has external validation but no kernel replay. Generator revision/seed is incompletely recorded. Novelty review covered primary arXiv texts, ancillaries, historical tables, citation paths, Exa and Crossref; Google Scholar returned HTTP 429, zbMATH and MathSciNet HTTP 403. A complete literature or forward-citation search is not claimed. Secondary novelty remains unresolved, including potential older linear covering constructions. No assertion is made that 1351 is optimal or that a 1350-word search has been exhausted.

10 Conclusion

The explicit object and two independent exhaustive algorithms support the eight stated upper bounds. The principal construction improves the verified 1475 bound found in this audit. The formal gap is precisely coverage replay for the frozen principal object, rather than construction recovery or executable coverage.

A Certificate references

Canonical files, metadata, original byte archives, hash files and reports reside under `research/covering-codes/certificates/`. The principal directory is `q7_n9_r4_m1351`; it also contains `witness.u16le.gz`, `witness-metadata.json` and `certificate-check.json`. The flat payload hash is

6f4425a72b66efa284e7ffc1bfd20117fc058e44bb46b37650f2c9dca5a72d2a.

`ARTIFACTS.sha256`, `artifact-search.md`, `adversarial-audit.md`, `certificate-design.md`, `formal-verification.json` and `literature.json` provide the audit chain. Earlier agents' stdout remains archival evidence only.

References

- [1] M. Marosi, New upper and lower bounds on covering codes $K_q(n, R)$ for alphabets of size $5 \leq q \leq 21$, arXiv:2608.19872v3, 2026.
- [2] G. Kéri, Tables for bounds on covering codes, November 2011 revision, <https://old.sztaki.hu/~keri/codes/index.htm>.
- [3] D. Gijswijt and S. Polak, Semidefinite lower bounds for covering codes, arXiv:2504.01932v2, 2026.
- [4] A. Florath, Covering-code formalization, arXiv:2606.09600, 2026; <https://github.com/florath/covering-codes-lean>, pinned commit as above.
- [5] P. R. J. Östergård, New constructions for q -ary covering codes, Ars Combinatoria 52 (1999), 51–63.